

Assignment 6: Basic Probability

Your name here

PSTAT 188

Contents

Instructions	1
STRATEGY: for getting started with probability problems	1
Exercise 1:	1
Exercise 2:	2
Exercise 3:	2
Exercise 4:	2

Instructions

This assignment does not require any code to be written. You can complete these theory problems by hand or you may type these up in latex, Rmd as per your choice.

You only need to upload a .PDF of a photo or scan of your handwritten work or typed up work.

STRATEGY: for getting started with probability problems

Working through probability problems can seem difficult at first. Use this strategy as you think critically about your approach to solve these problems.

- Write down events in words. Let A be the event
- Write down the probability you want to calculate in math notation using the events you defined. Like: Want to calculate $P(A \cap B)$
- Are these events independent? multiply probabilities or use multiplication rule
- Are these events mutually exclusive? add probabilities or use addition rule
- If the number of ways your event can unfold is long and complicated, look at the complement.

Once you have asked these questions, it'll be a matter of applying the correct formula.

Thinking critically to get answers to these questions is the challenging part of probability. With practice, this will get easier and become second nature!

Exercise 1:

- a. Draw 2 cards with replacement. What's the chance of getting two red cards?
- b. Draw 2 cards without replacement. What's the chance of getting two red cards?
- c. Draw 2 cards without replacement. What's the chance of getting the first red and second not red?

Exercise 2:

The table shows the results of a study in which researchers examined whether athletes participated in a special training program and whether they achieved above-average performance.

	Training Program	No Training	Total
Above Average	28	14	42
Average/Below	32	26	58
Total	60	40	100

Find the probability that...

- (a) ... an athlete has above-average performance, given the athlete participated in the training program.
- (b) ... an athlete participated in the training program.
- (c) ... an athlete has above-average performance, and participated in the training program.

Exercise 3:

A fair six sided die is thrown 4 times. Find the probability that a 5 is obtained each time.

Exercise 4:

Let X be a continuous random variable with density function

$$f(x) = \begin{cases} 3e^{-3x} & \text{if } x > 0 \\ 0 & \text{otherwise} \end{cases}$$

Find the probability that X takes values between 1 and 2.